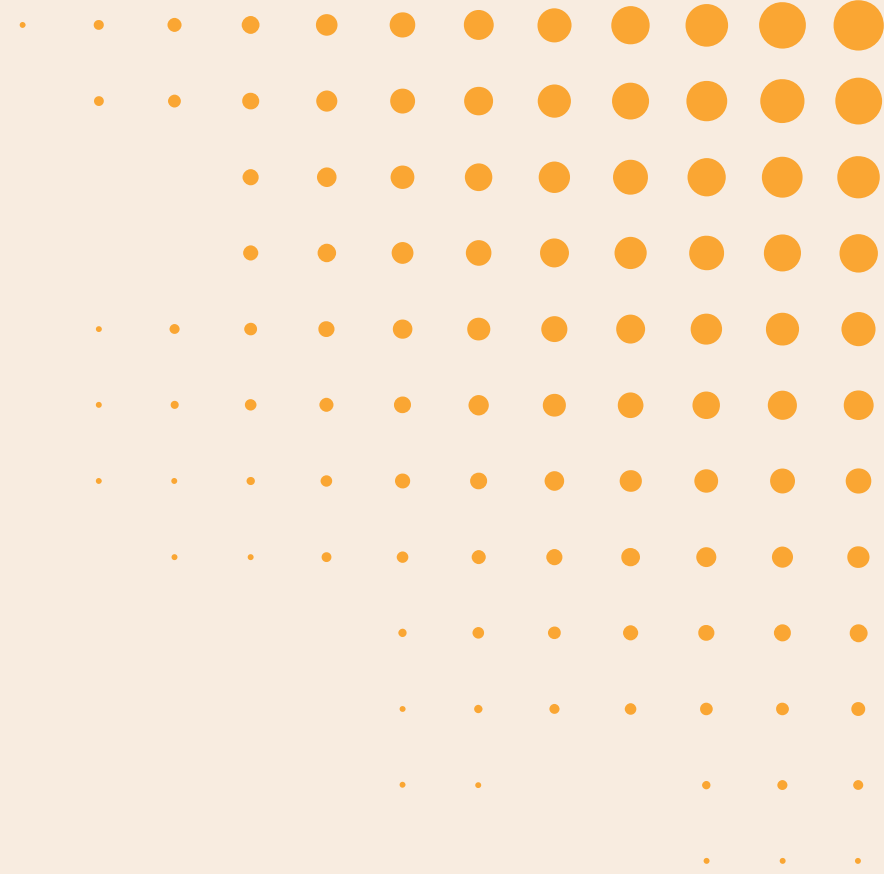
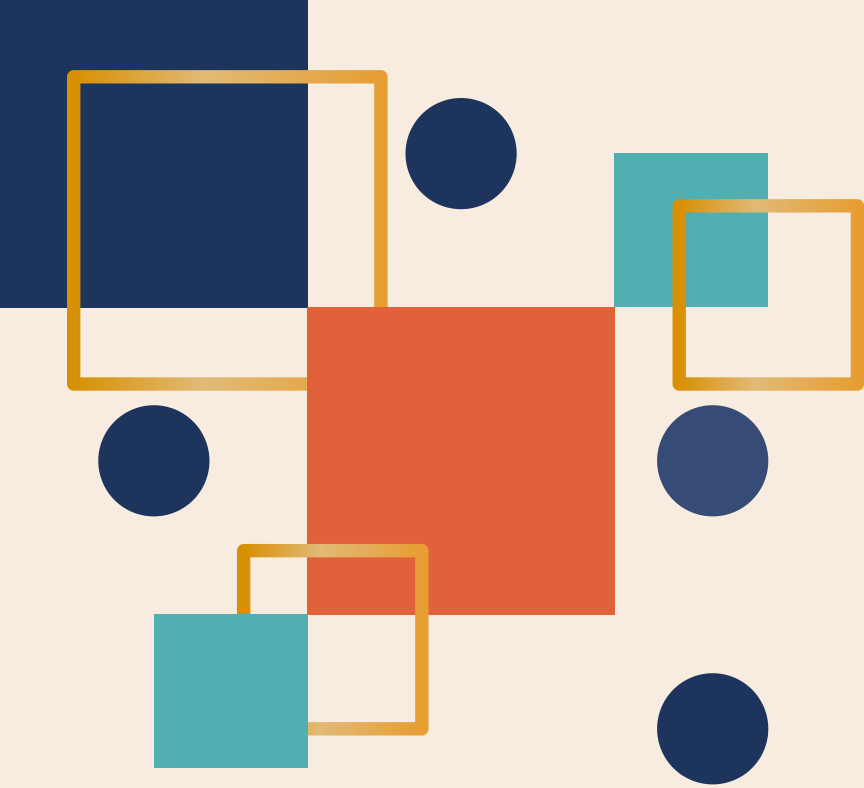




# AI-Generated vs Real Image Classification

Introdução à Aprendizagem Automática

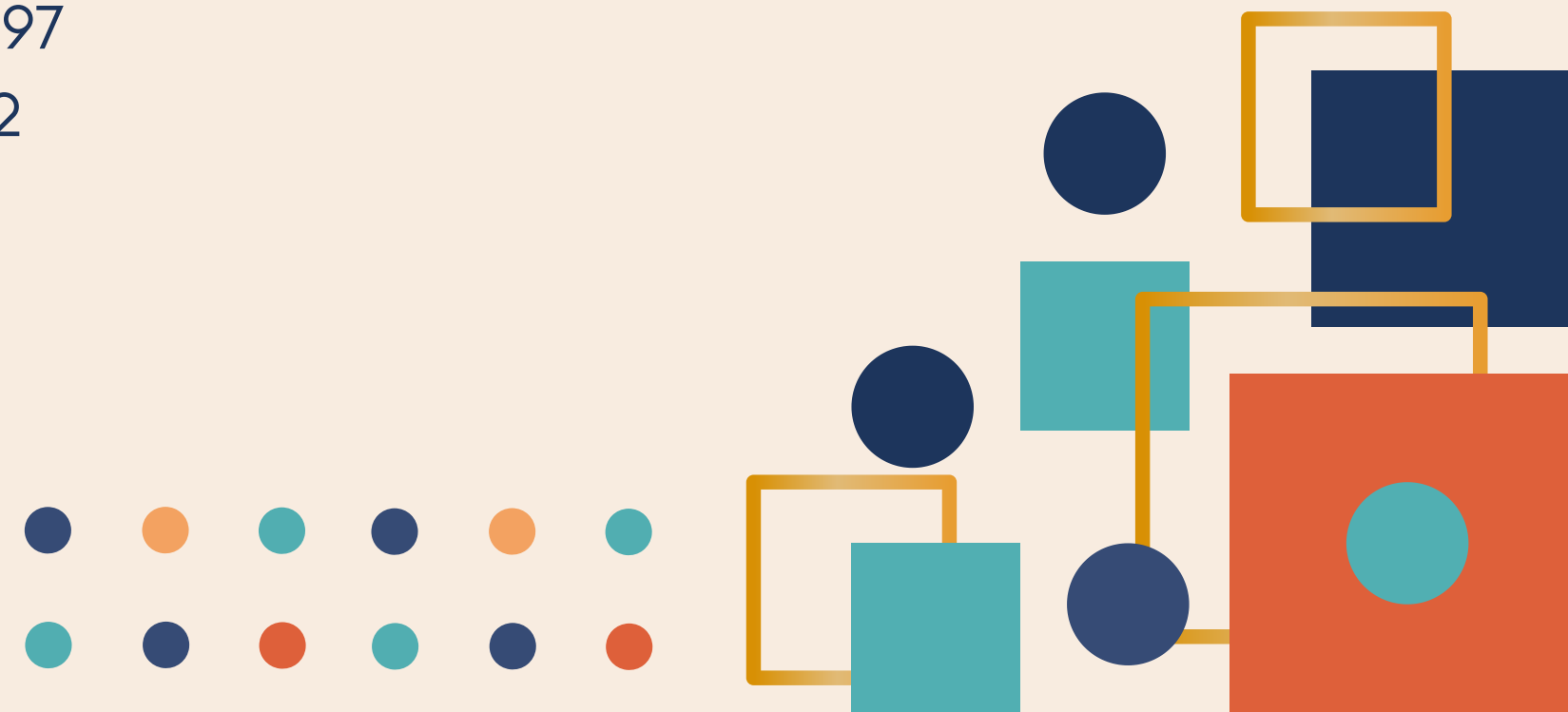
António Santos – 119139

Thiago Vicente – 121497

Miguel Neto – 119302

PL3

29/05/2026



# Problem

| Success rate (%) <sup>*</sup> |     |
|-------------------------------|-----|
| Users                         |     |
| Real images                   | 59% |
| AI generated images           | 63% |
| Overall                       | 62% |

*Is it safe to assume we can distinguish fantasy from reality?*

<sup>\*</sup> Excerpt from Figure 13, humans users, from:

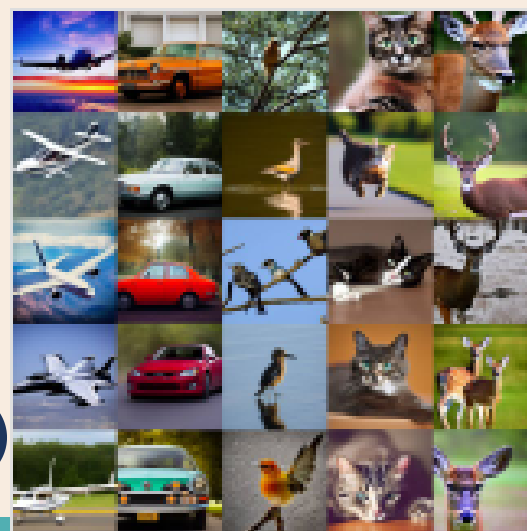
Thomas Roca, Anthony Cintron Roman, Jehú Torres Vega, Marcelo Duarte, Pengce Wang, Kevin White, Amit Misra, & Juan Lavista Ferres. (2025). How good are humans at detecting AI-generated images? Learnings from an experiment.

# Objectives

- **Classify images on their authenticity: *AI* or *REAL***
- **Surpass the current “natural” baseline**

# Dataset – CIFAKE

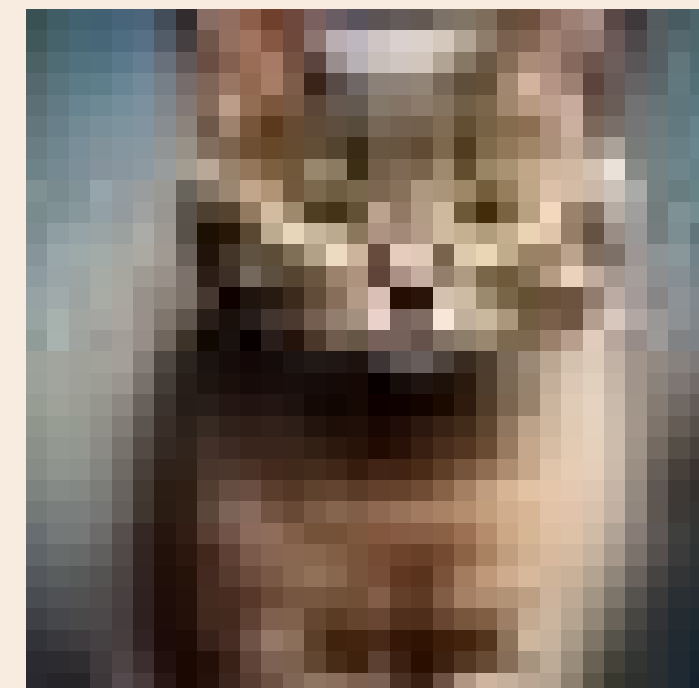
- Real images from **CIFAR-10**
- Generated images from **Stable Diffusion 1.4**
- **120000** total images with **32x32** resolution (**50-50 class split**)
  - **80000** for **training**
  - **20000** for **testing**
  - **20000** for **validation**



## CIFAKE: Real and AI-Generated Synthetic Images

Can Computer Vision detect when images have been generated by AI?

[k kaggle.com](https://www.kaggle.com)



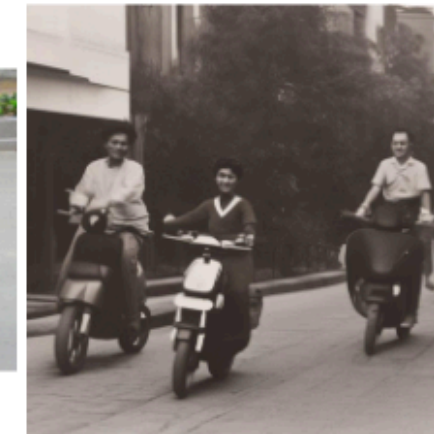
# Dataset – Defactify

- **Only for comparing performance**
- Real images from MS COCO
- Generated images from:
  - Stable Diffusion 2.1
  - Stable Diffusion XL
  - Stable Diffusion 3
  - DALL-E 3
  - MidJourney v6

Caption: "Two men riding mopeds, one with a woman and boy riding along."



Real (MS COCO)



Stable Diffusion 2.1



Stable Diffusion XL



Stable Diffusion 3



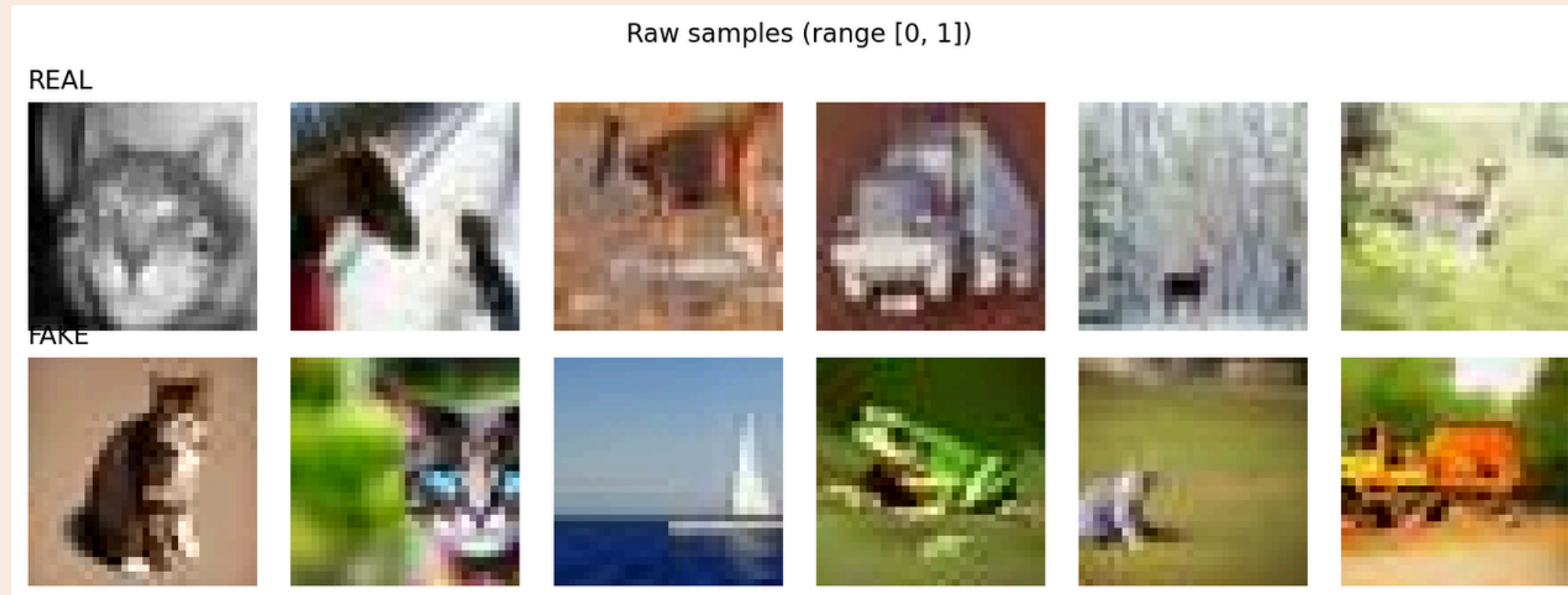
DALL-E 3



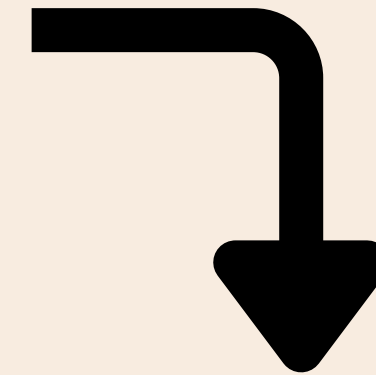
MidJourney v6



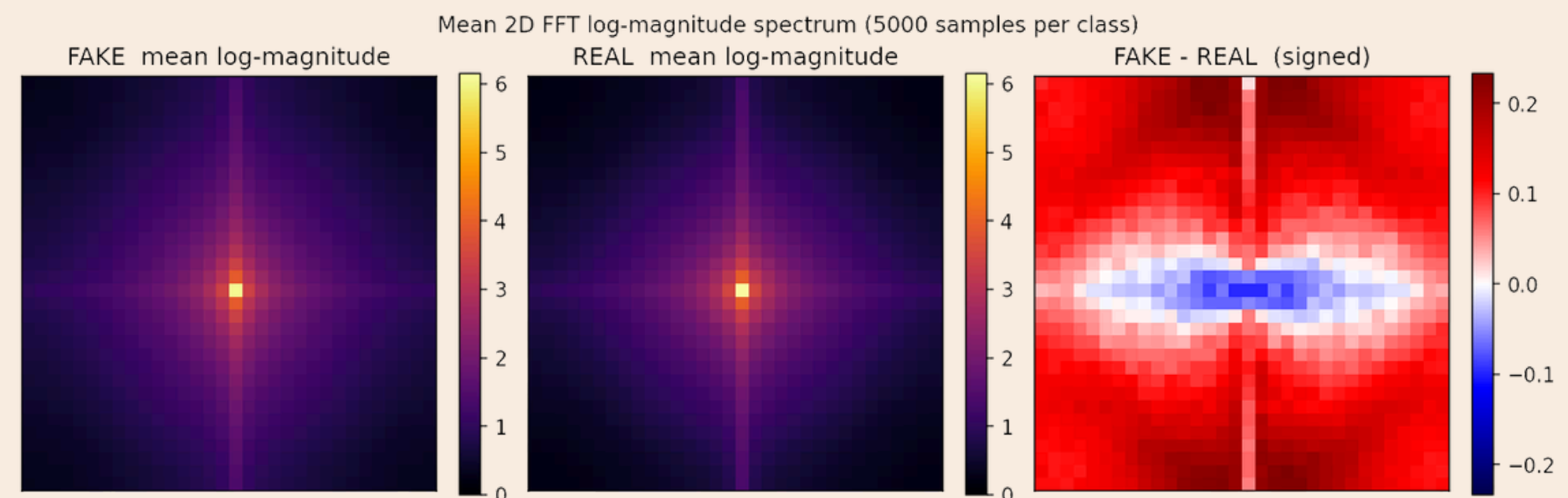
# Dataset - Analysis (CIFAKE)



Not visually distinct



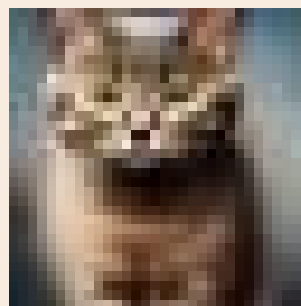
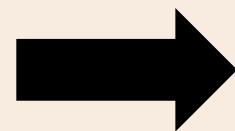
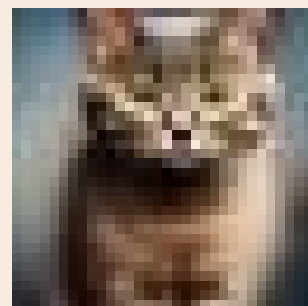
FFT log-magnitude spectrum



# Preprocessing

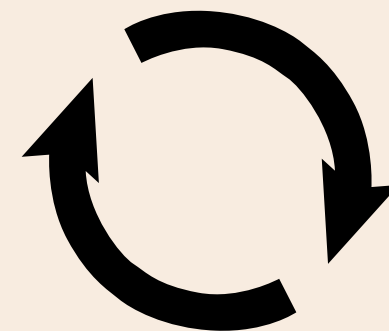
## Augmentation:

- Mirroring
- Small rotations ( $\pm 10^\circ$ )
- On the fly per epoch



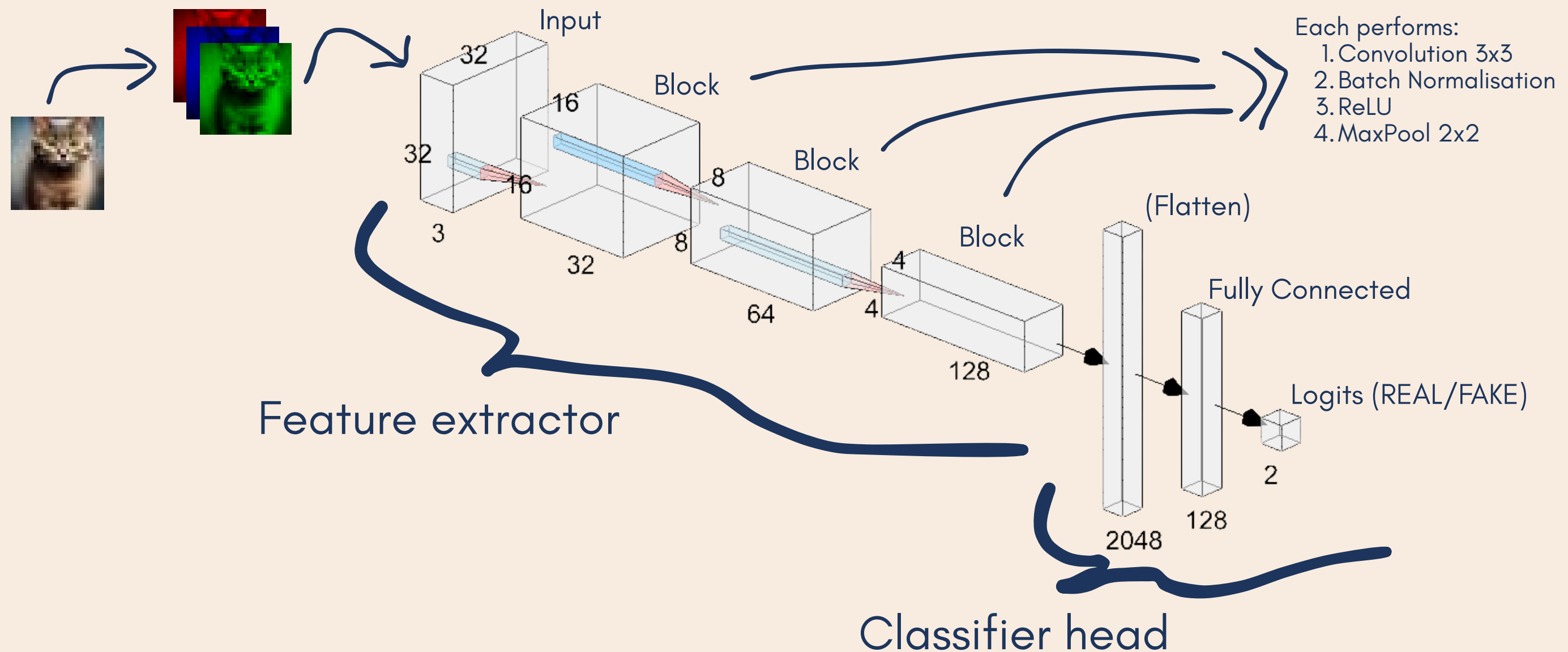
## Scaling:

- Z-score standardization
- Per each color channel
- Global  $\mu$  and  $\sigma$



# Model Analysis and Selection

## First candidate - Convolutional Neural Network (CNN)

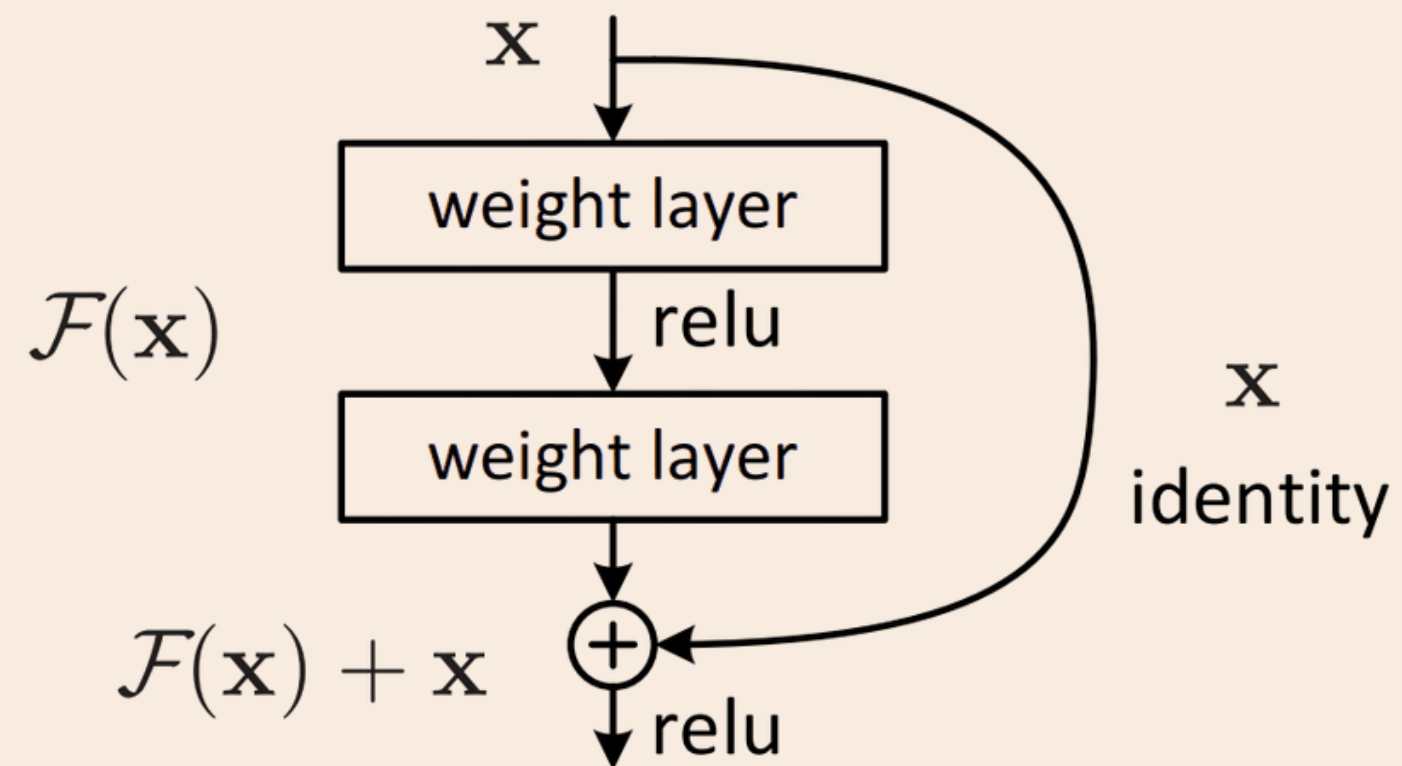


Visualisation generated on <https://alexlenail.me/NN-SVG/AlexNet.html>  
based on the implementation for this project



# Model Analysis and Selection

## Second candidate - Residual Network (ResNet)

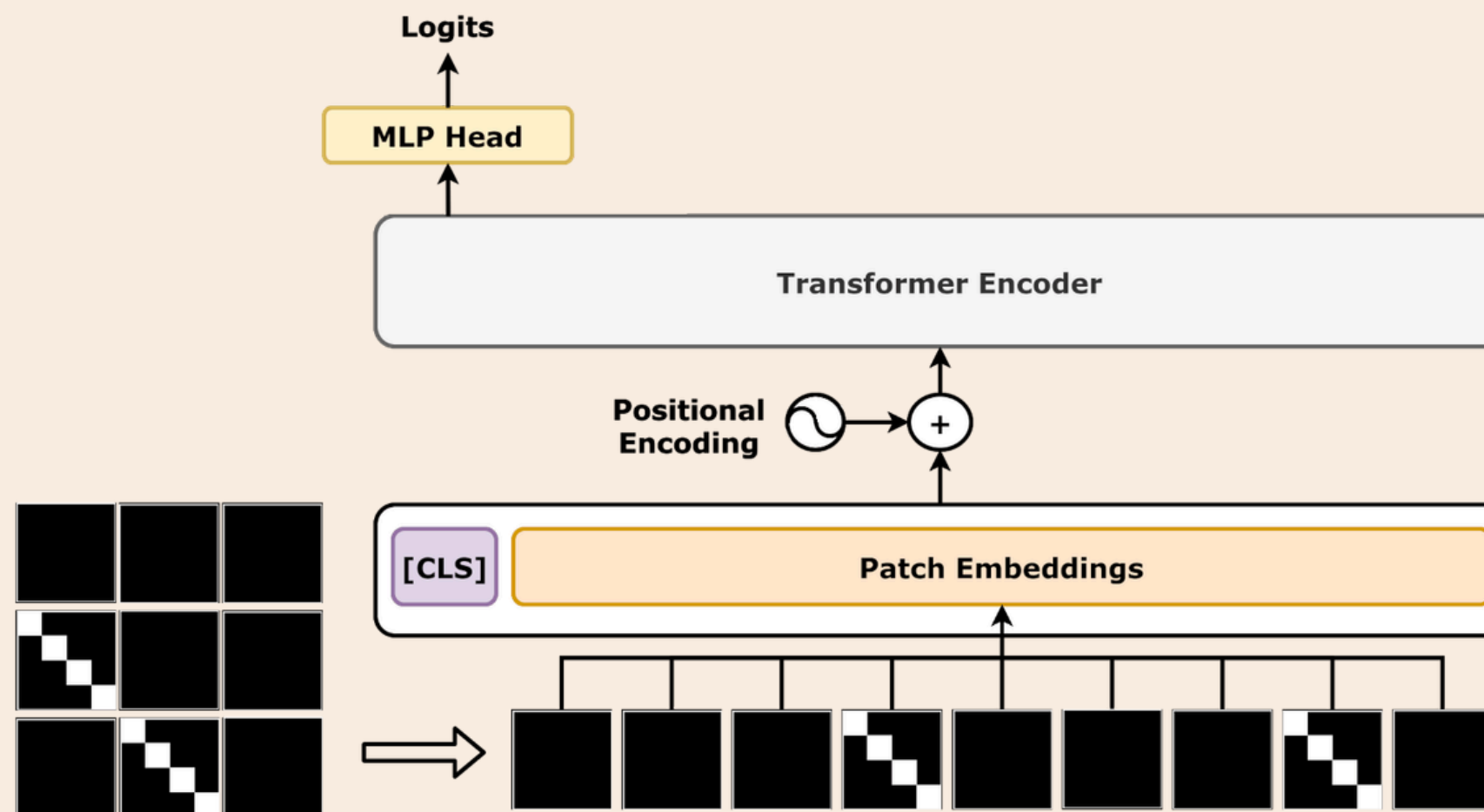


Adjustments applied upon selection:

- **Stem:**  $3 \times 3$  stride-1, no maxpool
  - keeps  $32 \times 32$
- **Output Layer:** 2 logits
- **No** pre-trained weights

# Model Analysis and Selection

## Third candidate - Vision Transformer (ViT)



- Requires upscaling to 224x224
- Data-hungry
  - Dataset is too small
- Hardly reproducible on conventional hardware
  - 80M+ parameters!
- Pretrained (From external data)

# Model Analysis and Selection

## Selected Models

Due to resource and data *constraints*:

**CNN**

**ResNet-18**

$3 \times 32 \times 32 = 3072$  input features

matplotlib

PyTorch

scikit  
learn

# Hyper-parameter selection

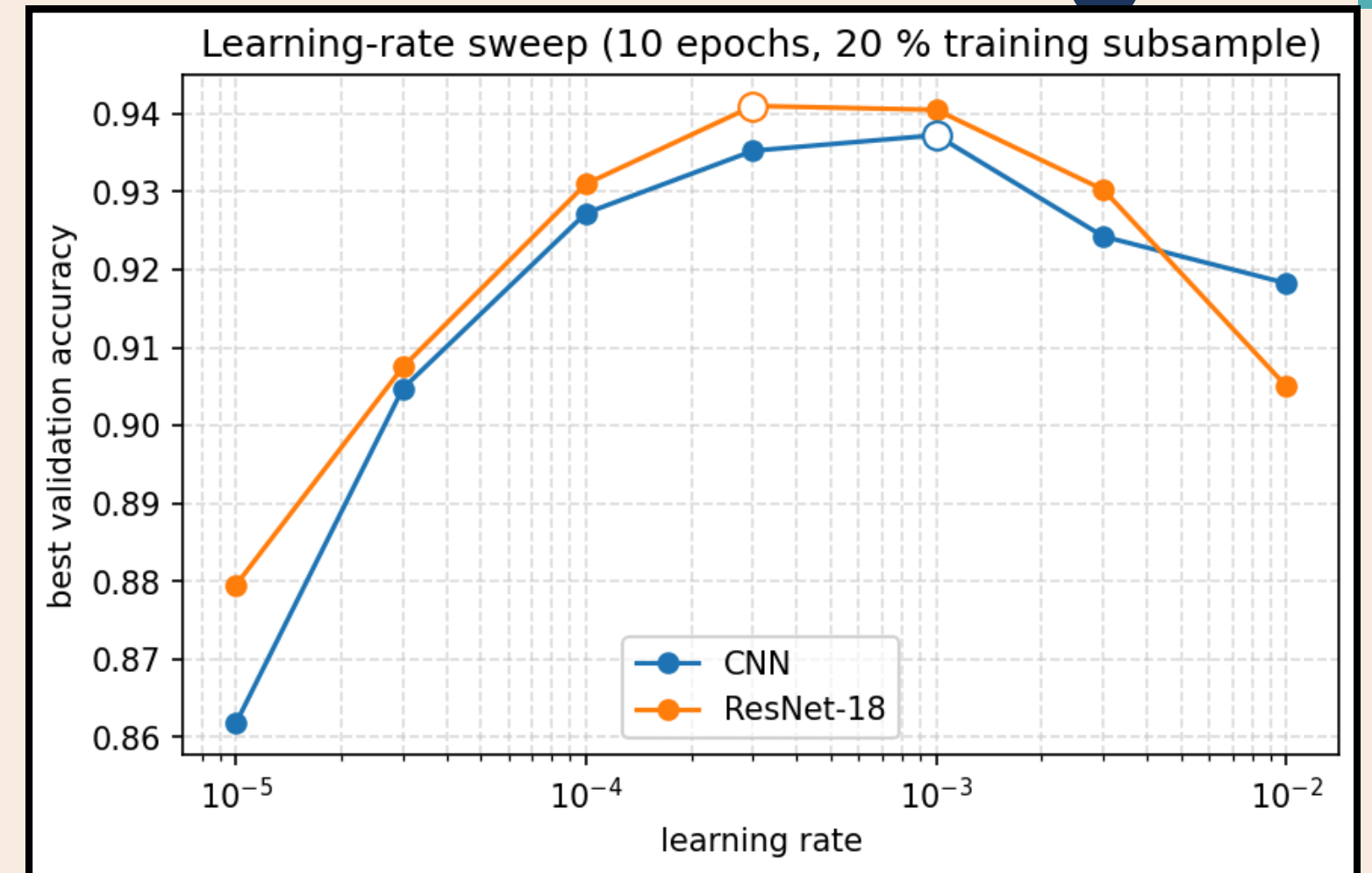
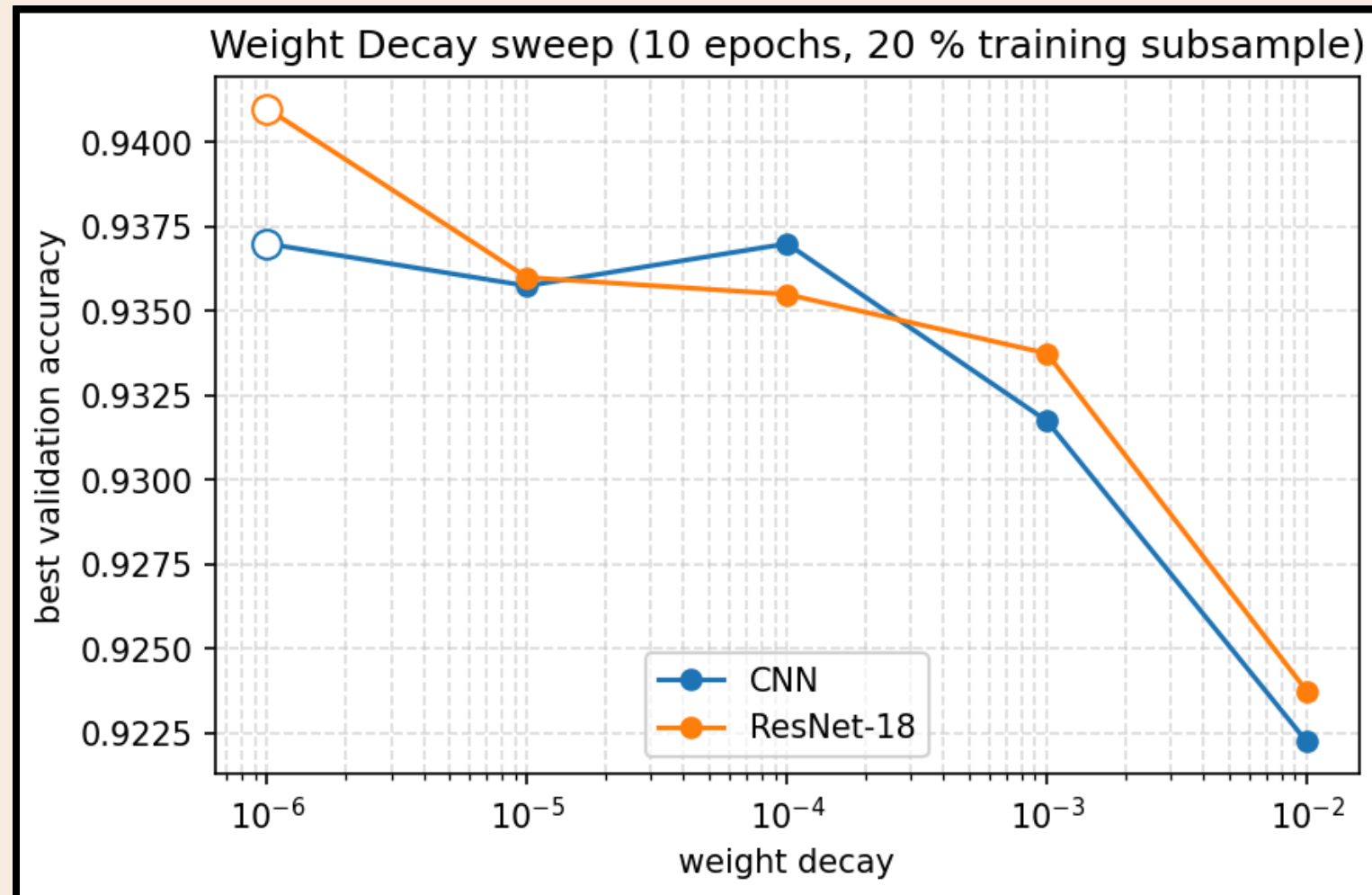
## What we swept

- Learning rate
- Weight decay

## Protocol

- 4 orders of magnitude
- 10 epochs
- 20% training subsample
- Pick max val-acc per model

# Hyper-parameter selection



| Best values... | Weight Decay | Learning rate |
|----------------|--------------|---------------|
| CNN:           | 1e-4         | 1e-3          |
| ResNet         | 1e-6         | 3e-4          |



# Training

## Setup ...

- Adam optimizer
- 256 batch size
- Cross Entropy loss
- Seed 42
- Weight decay (from previous slide)

## Early abort ...

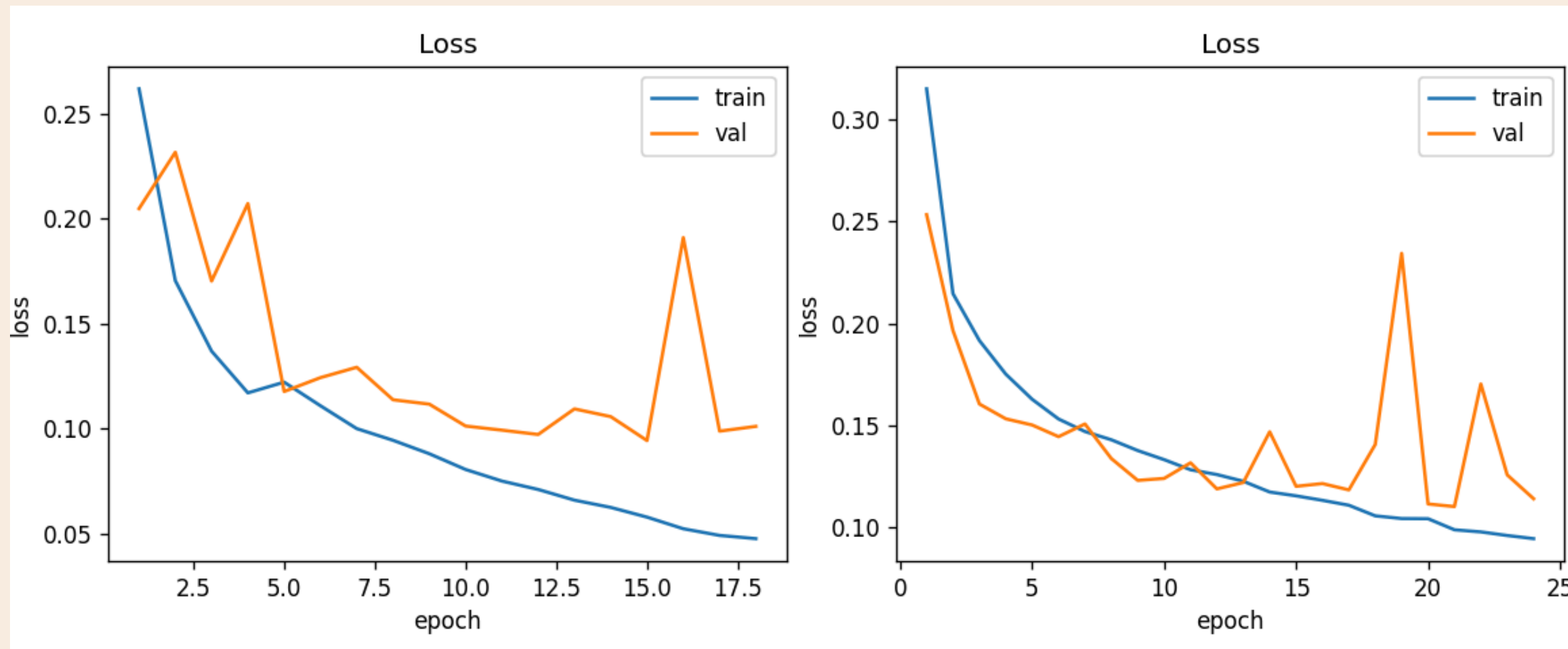
- no fixed epoch cap
- patience = 3

## Checkpoints ...

- Best val-acc weights
- not last epoch



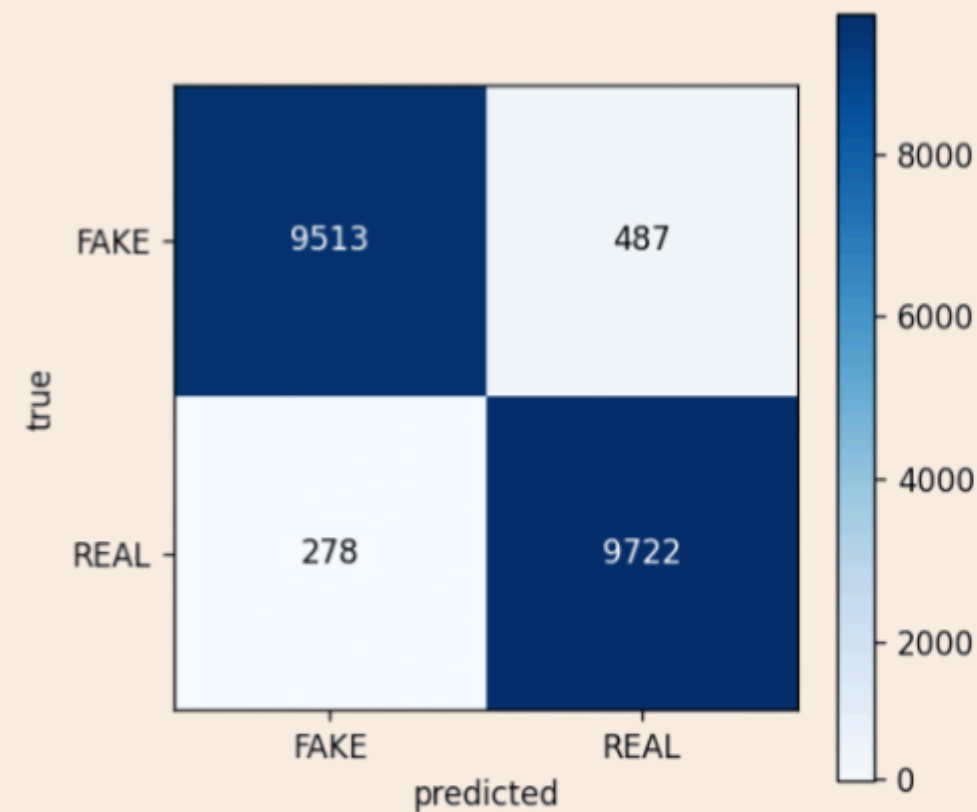
# Training



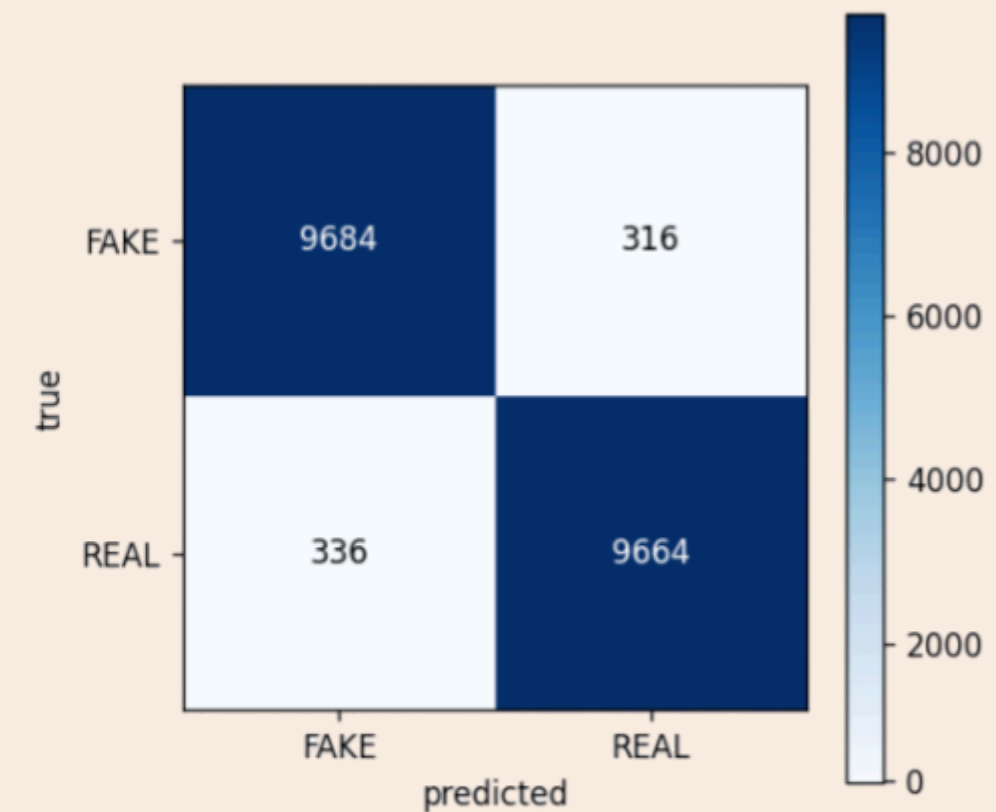
↑ **ResNet**

↑ **CNN**

# Discussion and Results



(a) CNN



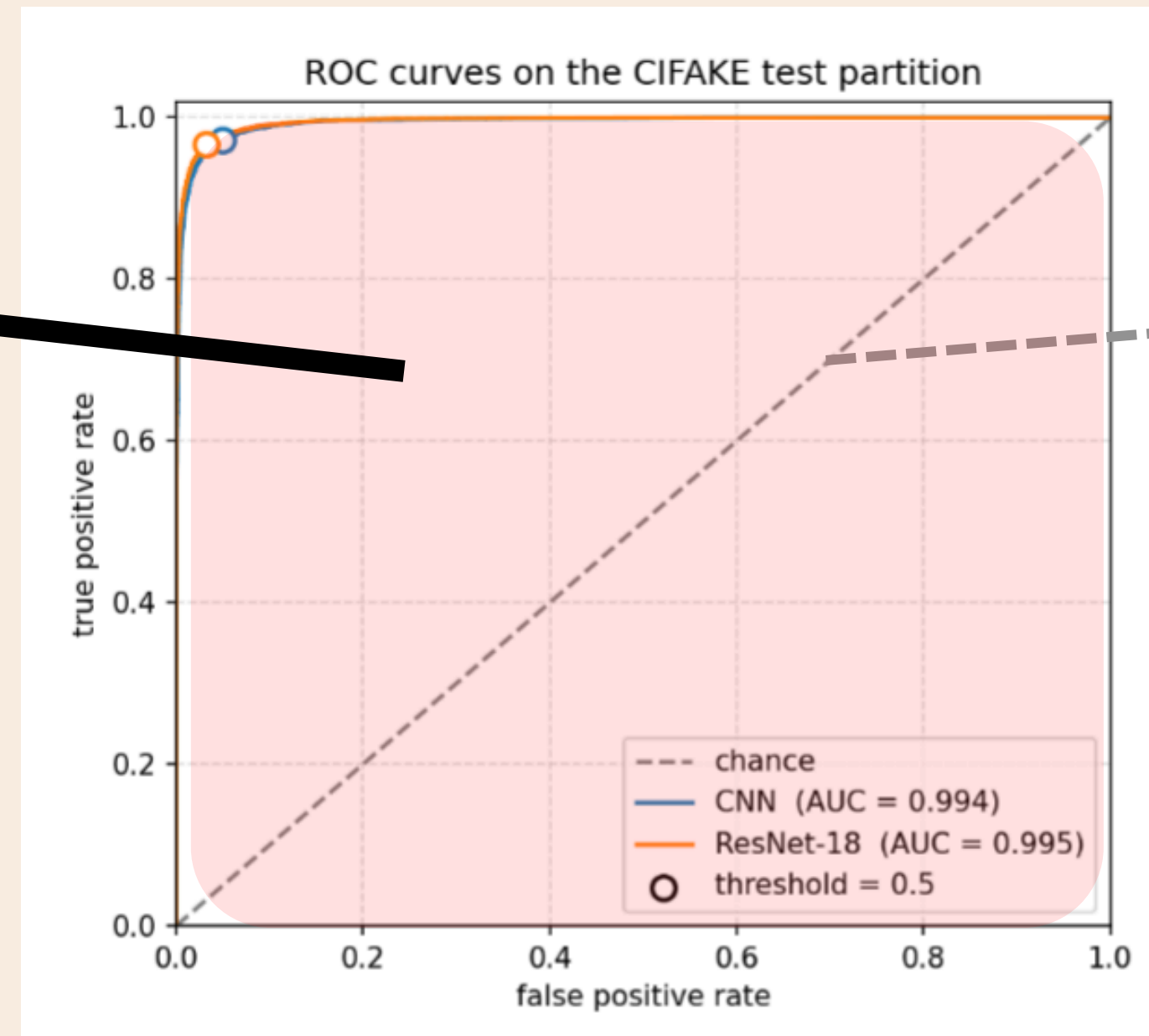
(b) ResNet-18

Only on CIFAKE!

| Model              | Acc.         | Prec.        | Rec.         | F1           | AUC          | Time (s) |
|--------------------|--------------|--------------|--------------|--------------|--------------|----------|
| CNN (0.36 M)       | 0.962        | 0.952        | <b>0.972</b> | 0.962        | 0.994        | 94       |
| ResNet-18 (11.2 M) | <b>0.967</b> | <b>0.968</b> | 0.966        | <b>0.967</b> | <b>0.995</b> | 377      |

# Discussion and Results

AUC



Random classifier

# Discussion and Results

## Effect of normalization

| Model (norm)       | Acc.         | Prec.        | Rec.         | F1           | AUC          |
|--------------------|--------------|--------------|--------------|--------------|--------------|
| CNN, with          | <b>0.962</b> | <b>0.952</b> | <b>0.972</b> | <b>0.962</b> | <b>0.994</b> |
| CNN, without       | 0.926        | 0.894        | 0.966        | 0.929        | 0.983        |
| ResNet-18, with    | <b>0.967</b> | 0.968        | <b>0.966</b> | <b>0.967</b> | <b>0.995</b> |
| ResNet-18, without | 0.966        | <b>0.970</b> | 0.962        | 0.966        | <b>0.995</b> |

# Discussion and Results

Defactify dataset!

## Validation on other generators

SD



| Generator     | Model     | Acc          | Prec         | Rec          | F1           | AUC          |
|---------------|-----------|--------------|--------------|--------------|--------------|--------------|
| SD 2.1        | CNN       | 0.560        | 0.535        | <b>0.912</b> | <b>0.675</b> | <b>0.686</b> |
|               | ResNet-18 | <b>0.566</b> | <b>0.543</b> | 0.828        | 0.656        | 0.634        |
| SDXL          | CNN       | 0.537        | 0.521        | <b>0.932</b> | <b>0.668</b> | 0.626        |
|               | ResNet-18 | <b>0.566</b> | <b>0.545</b> | 0.806        | 0.650        | <b>0.630</b> |
| SD 3          | CNN       | <b>0.671</b> | <b>0.613</b> | <b>0.928</b> | <b>0.738</b> | <b>0.799</b> |
|               | ResNet-18 | 0.614        | 0.581        | 0.822        | 0.680        | 0.675        |
| DALL-E 3      | CNN       | 0.428        | 0.460        | <b>0.828</b> | 0.591        | 0.457        |
|               | ResNet-18 | <b>0.464</b> | <b>0.478</b> | 0.792        | <b>0.596</b> | <b>0.487</b> |
| Midjourney v6 | CNN       | 0.449        | 0.472        | <b>0.846</b> | <b>0.606</b> | 0.446        |
|               | ResNet-18 | <b>0.467</b> | <b>0.480</b> | 0.784        | 0.595        | <b>0.538</b> |



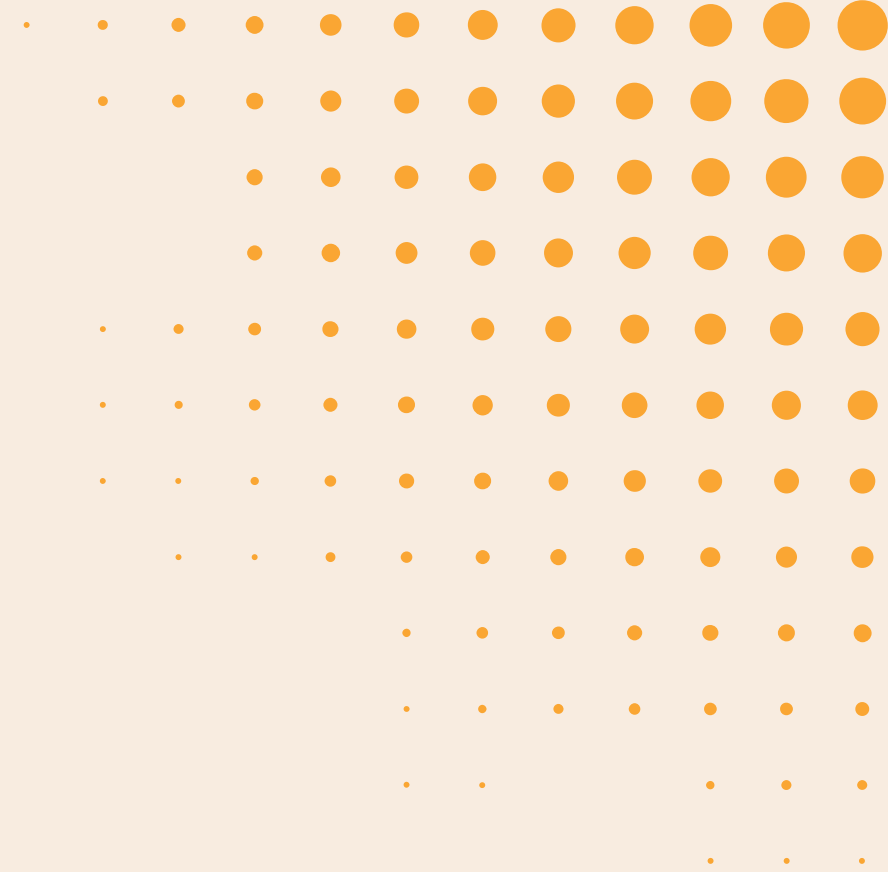
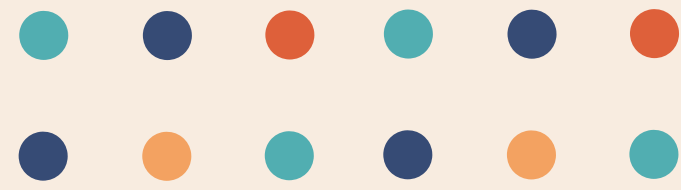
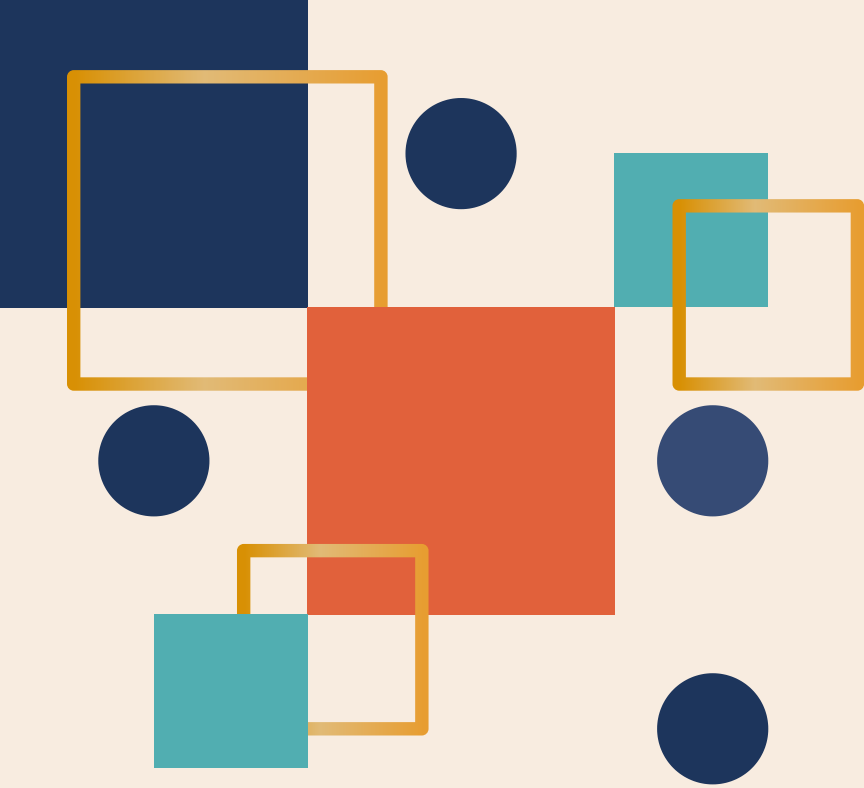
# Conclusion

- **The model learned** Stable Diffusion 1.4 adequately.
- Performance fell off for other generators.
- Model tends to REAL.
- Model didn't generalize → shouldn't be used in production.

# Reflection

- Every generator **has its own signature**
- A detector trained on a model/signature **might** only learn that one
- As generators keep improving, the **detectors must be retrained and maintained.**





OBRIGADO!

